

DTU



Lazaros Nalpantidis

3D Point Cloud Processing - Clustering & Regression



Outline

- What are Clustering and Regression? A taxonomy of ML algorithms
- Why is ML important for Perception of Autonomous Systems?
- Regression
- Clustering
 - k-means
 - Mean Shift
 - DBSCAN
 - Hierarchical Clustering
- Summary



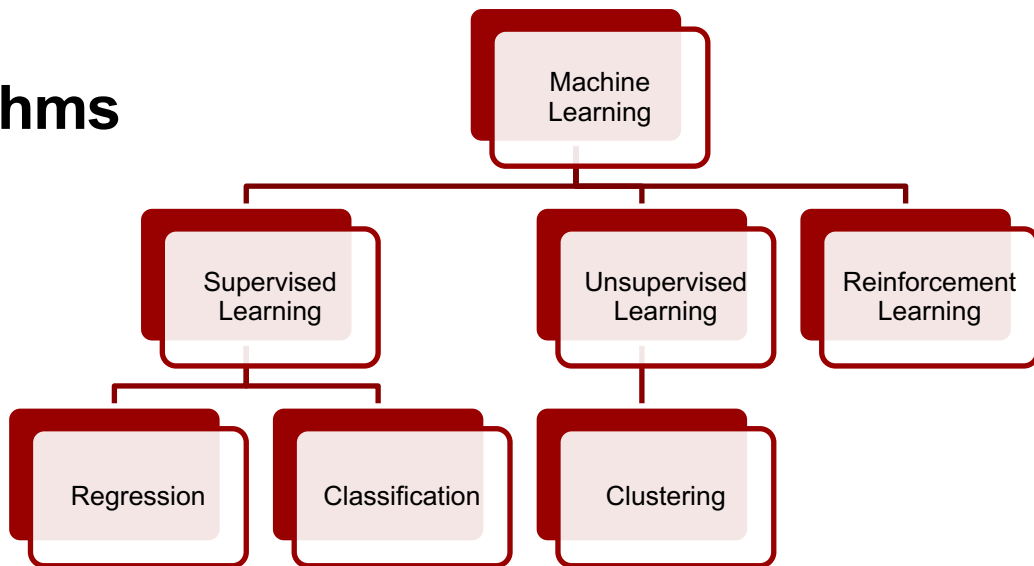
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What are Clustering and Regression?

A taxonomy of ML algorithms



- **Supervised Learning** – The target values are known
 - Regression – The target value is numeric
 - Classification – The target value is nominal
- **Unsupervised Learning** – The target values are unknown
 - Clustering – Group together similar instances
- **Reinforcement Learning** – Interacting with a dynamic environment the system must perform a certain goal.

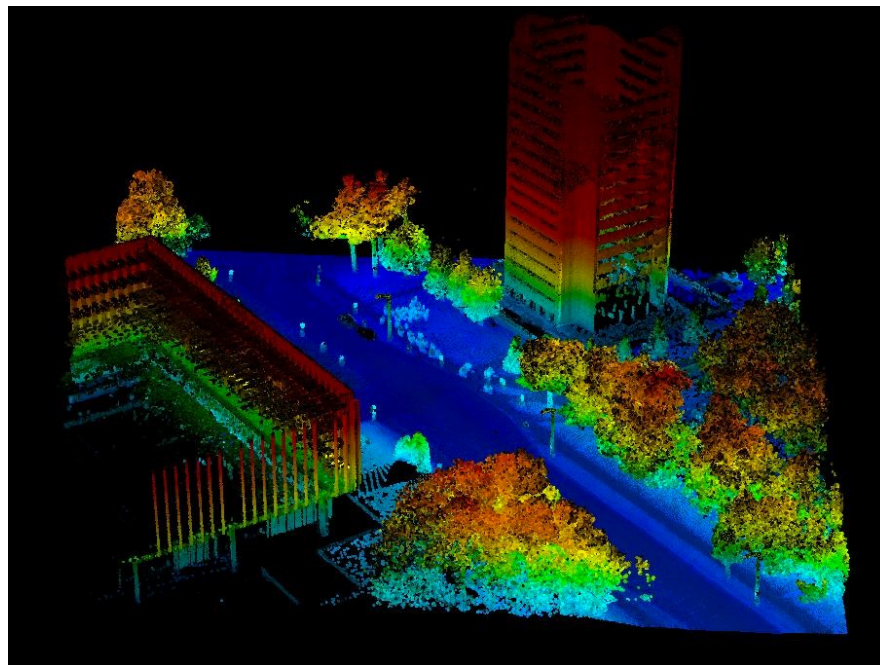


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Why is ML important for Perception of Autonomous Systems?

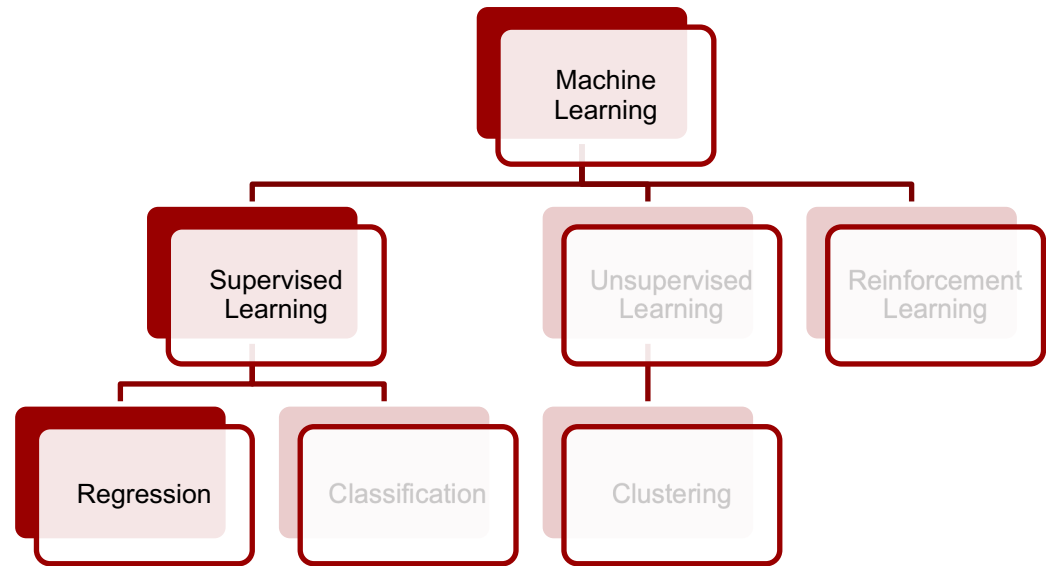
- Create initial object hypotheses
 - input to pose estimation?
- Abstract representation of scene
- ...what else?





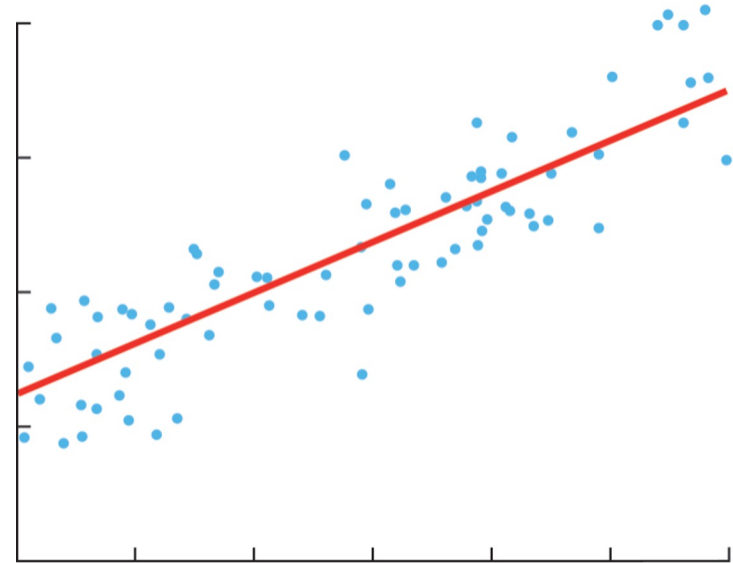
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- **Supervised Learning** – The target values are known
 - Regression – The target value is numeric

- Regression tries to assign correct significance (weights) to input variables and formulate a weighted linear combination of them that can predict the output variables.
- The number of input and output variables can vary depending on the specific problem.
- The final output of the algorithm is a regression line



- The goal is to find an equation $f(\cdot)$ that models the relationship between input x and output y such as:

$$y = f(x) + \varepsilon$$

- Regression Models:
 - Linear Regression: Assumes that function f is linear.
 - Locally Weighted Regression: Creates multiple linear models on small neighbor data.

- The performance for the linear regression method is evaluated by an error function.
- The most used is the Mean Squared Error (MSE) :

$$MSE = \frac{1}{N} \sum_{n=1}^N (t - y)^2$$

...where t is the true value of the learned output and y is the predicted value.

- **Linear Regression**

- Linear Regression assumes that $f(x)$ is a linear combination between the inputs x and a set of regression coefficients b :

$$y = f(b, x) = b_1 x_1 + b_2 x_2 + b_3 x_3 + \dots + b_n x_n + c = b^T x + c$$

- The goal of linear regression is to find regression coefficients b that minimize the squared error between the true values of the output and the predicted values.

- **Polynomial Regression**

- In the case that the mapping between the inputs x and the output is not expressed well by a straight line we can use polynomial regression.
- This is done by adding dimensions to the input data, where n is the degree of polynomial.

$$X = [x, x^2, x^3, \dots, x^n]$$

- The degree of polynomial can be found by cross-validation.



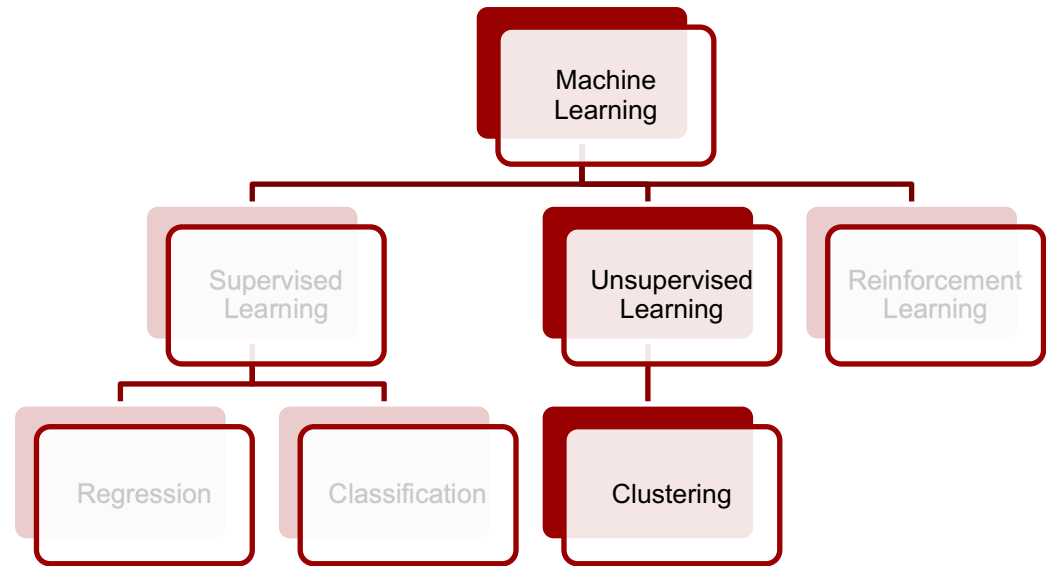
Regression

- Can we use Regression on our 3D point clouds?
- What problems could we tackle?
 - Line fitting
 - Plane fitting
 - Fitting other (non-linear) surfaces
 - ...what else?



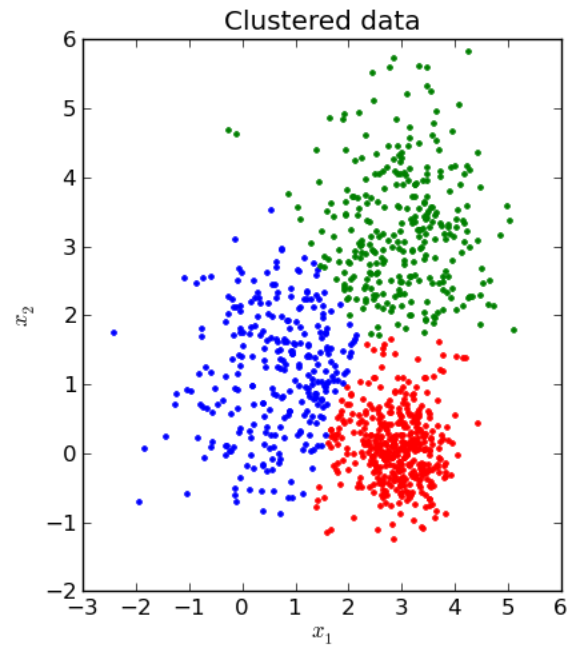
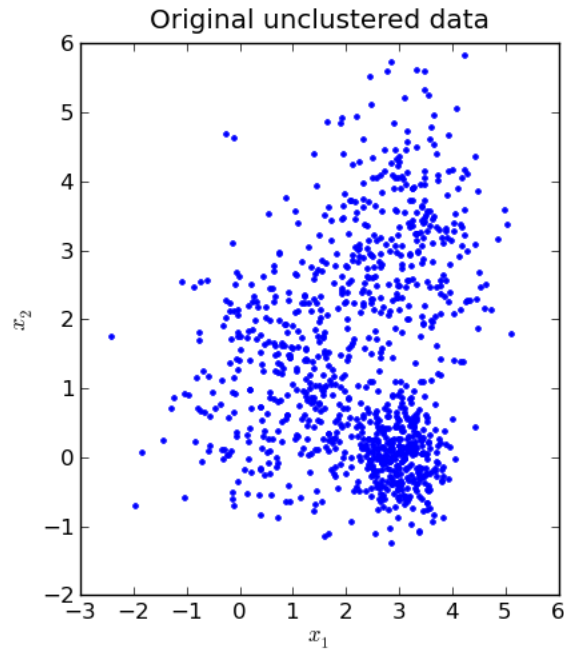
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- **Unsupervised Learning** – The target values are unknown
 - Clustering – Group together similar instances

- Partitions a point cloud into groups of (similar) points, i.e. clusters,
- Finds the underlying structure of the point cloud.



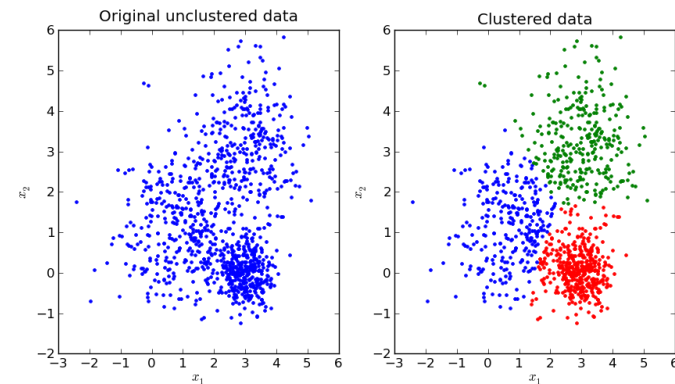
Clustering – k-means

- Partitions a point cloud into k clusters, such that each point belongs to one cluster
- Assigns point to clusters, so as to minimize the Sum of Squared Euclidean Distances (Distortion) between points and their assigned cluster centers.

$$J = \sum_k \sum_{n \in k} d(x_n, \mu_k)^2$$

...where x is a vector representing the n^{th} data point assigned to cluster k and μ_k is the centroid the cluster k .
The function $d()$ is the Euclidean distance between x and μ .

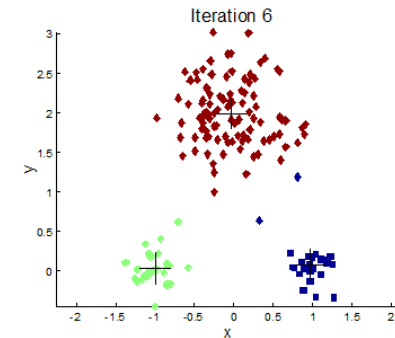
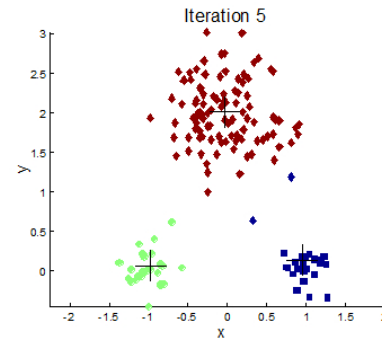
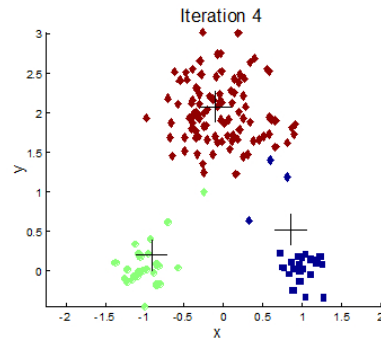
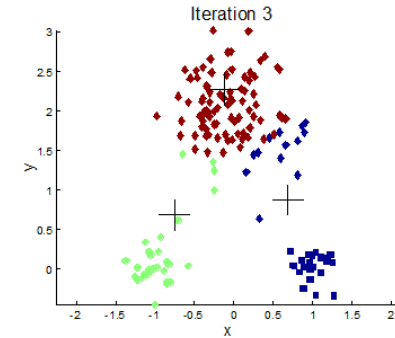
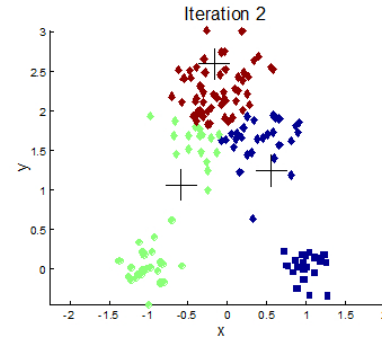
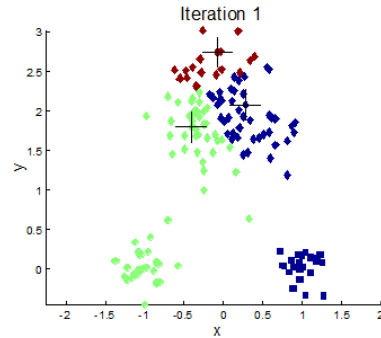
- **Important:** K-means requires known number of clusters k



Clustering – k-means

It is an iterative process:

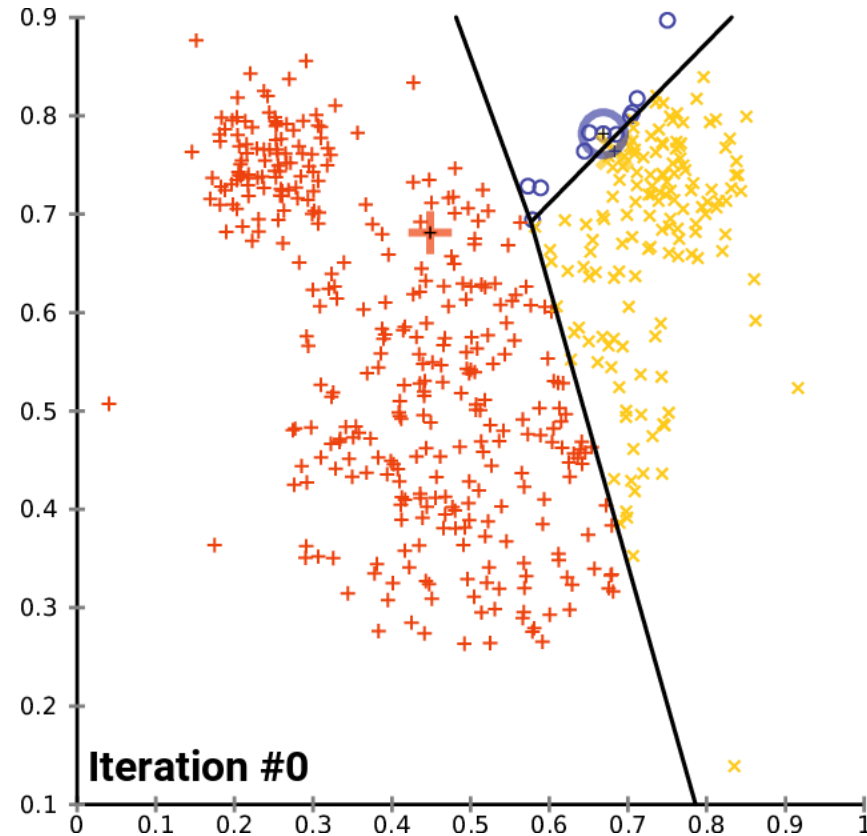
- Step 1:
 - Randomly initialize k cluster centers
- Step 2:
 - Assign each point to its nearest cluster center
- Step 3:
 - Re-compute each cluster-center as the centroid of all points assigned to each cluster
- Step 4:
 - Check convergence/stop criterion, otherwise repeat Steps 2-4



Clustering – k-means

It is an iterative process:

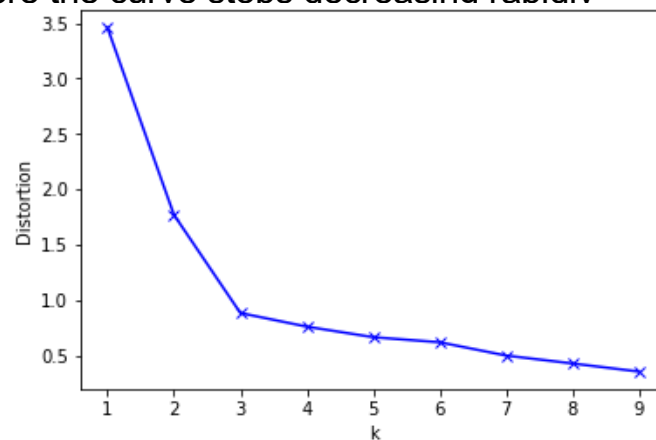
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How to define k ?

– Elbow method

- Run k-means for several k and calculate distortion for each k
 - » Distortion: sum of squared distances of each point to the center of the closest cluster
- Plot distortion vs k
- Look for k where the curve stops decreasing rapidly



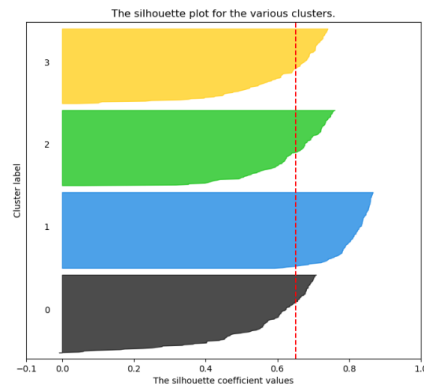
Clustering – k-means

How to define k ?

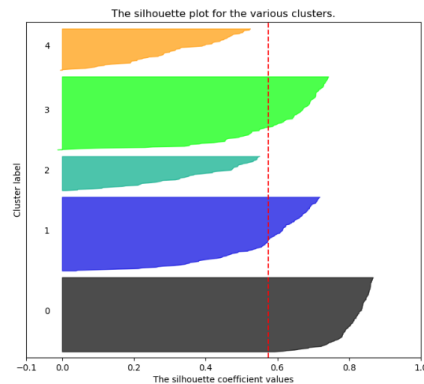
– Silhouette Analysis

- Calculate Silhouettes for various k
 - Silhouette coefficient shows how far each sample is from other cluster centers
 - » +1: far from others
 - » 0: at the boundary
 - » -1: sample is on average closer to the points of another cluster
- Thickness shows cluster size (number of points assigned to cluster)
- Avoid k that leads to:
 - Scores for clusters < average
 - Individual scores < 0
 - Big differences in cluster sizes (thickness)

Silhouette analysis for KMeans clustering on sample data with $n_clusters = 4$



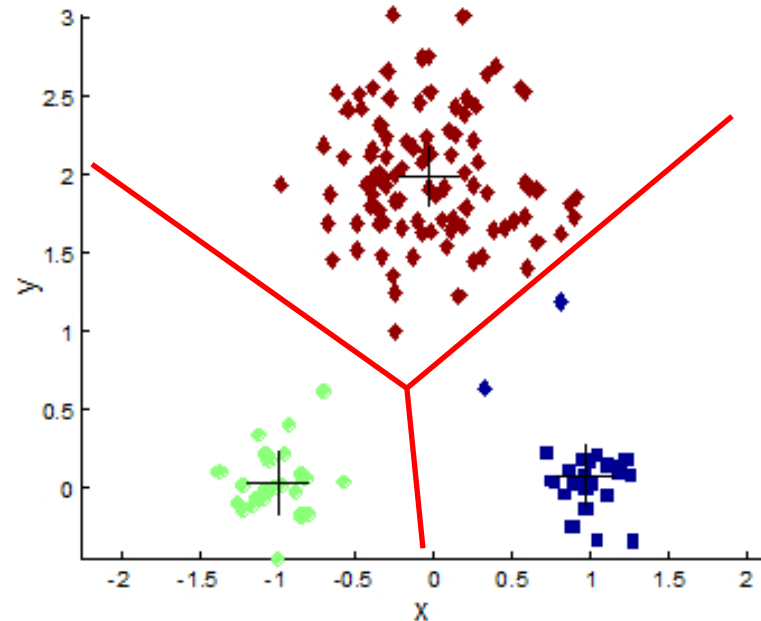
Silhouette analysis for KMeans clustering on sample data with $n_clusters = 5$



Clustering – k-means

Issues

- Need to define k
- Final cluster assignments depend on initialization
 - » Cluster assignments may be different on different runs
 - » K-means may not achieve the global optimum
- Can describe only convex cluster geometries
- Computationally demanding as the number of points and dimensions increase



3D Point Cloud Segmentation by k-means

Which space to apply Clustering? Is it your point cloud's 3D space?

What should be the dimensionality?

- dimensionality = 3 ? $\rightarrow$ X, Y, Z
- dimensionality = 6 ? $\rightarrow$ X, Y, Z, R, G, B
- dimensionality = 6 ? $\rightarrow$ X, Y, Z, L, U, V
- dimensionality = 6 ? $\rightarrow$ X, Y, Z, H, S, V
- dimensionality > 6 ? $\rightarrow$ X, Y, Z, H, S, V, gradients, texture, ... ??

Clustering – Mean Shift

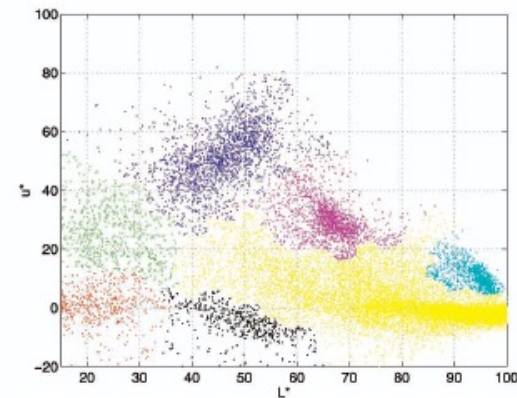
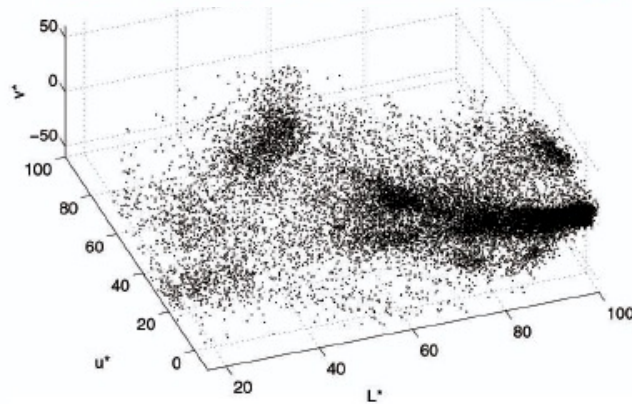
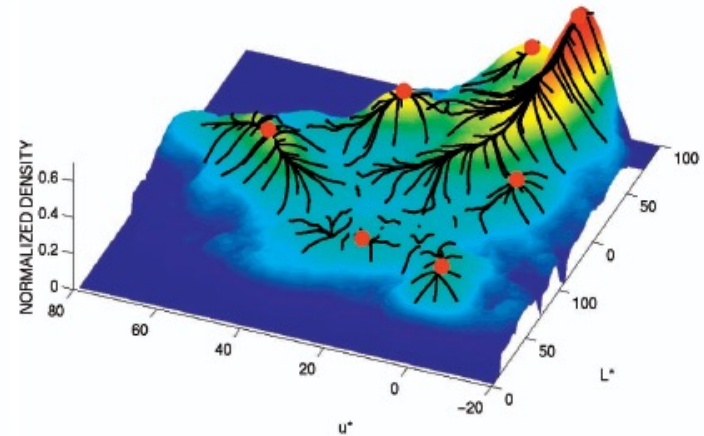
- **Mean Shift** is an algorithm that finds the maxima—the modes—of a density function given discrete data sampled from that function. Thus, it is a density hill climbing algorithm.
 - Every point gives rise to a cluster.
 - Each cluster is defined by the radius, a.k.a. “bandwidth”, h of its region, and a kernel function K that is used to calculate the contribution of the points included within this radius.
 - In the end, only unique clusters are considered.
- So, we do not need to set the number of clusters!
 - However, we need to define the bandwidth (and the kernel function).

Dorin Comaniciu and Peter Meer, “Mean Shift: A robust approach toward feature space analysis,” IEEE Transactions on Pattern Analysis and Machine Intelligence. 2002. pp. 603-619.

Yizong Cheng, “Mean shift, mode seeking, and clustering,” IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 17, pp. 790–799, Aug 1995.

Mean shift algorithm

- Try to find *modes* of this non-parametric density



Kernel density estimation

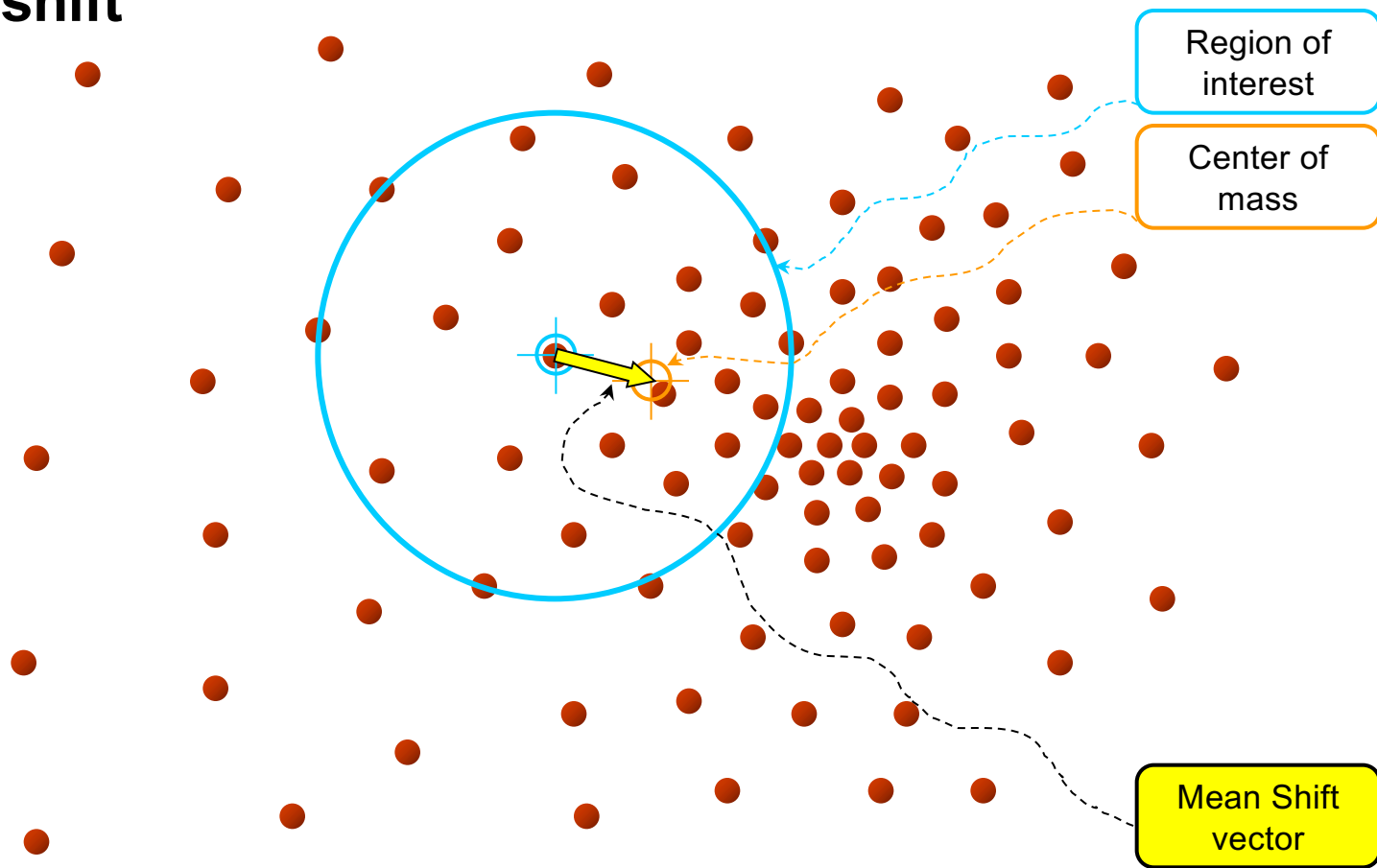
Kernel density estimation function

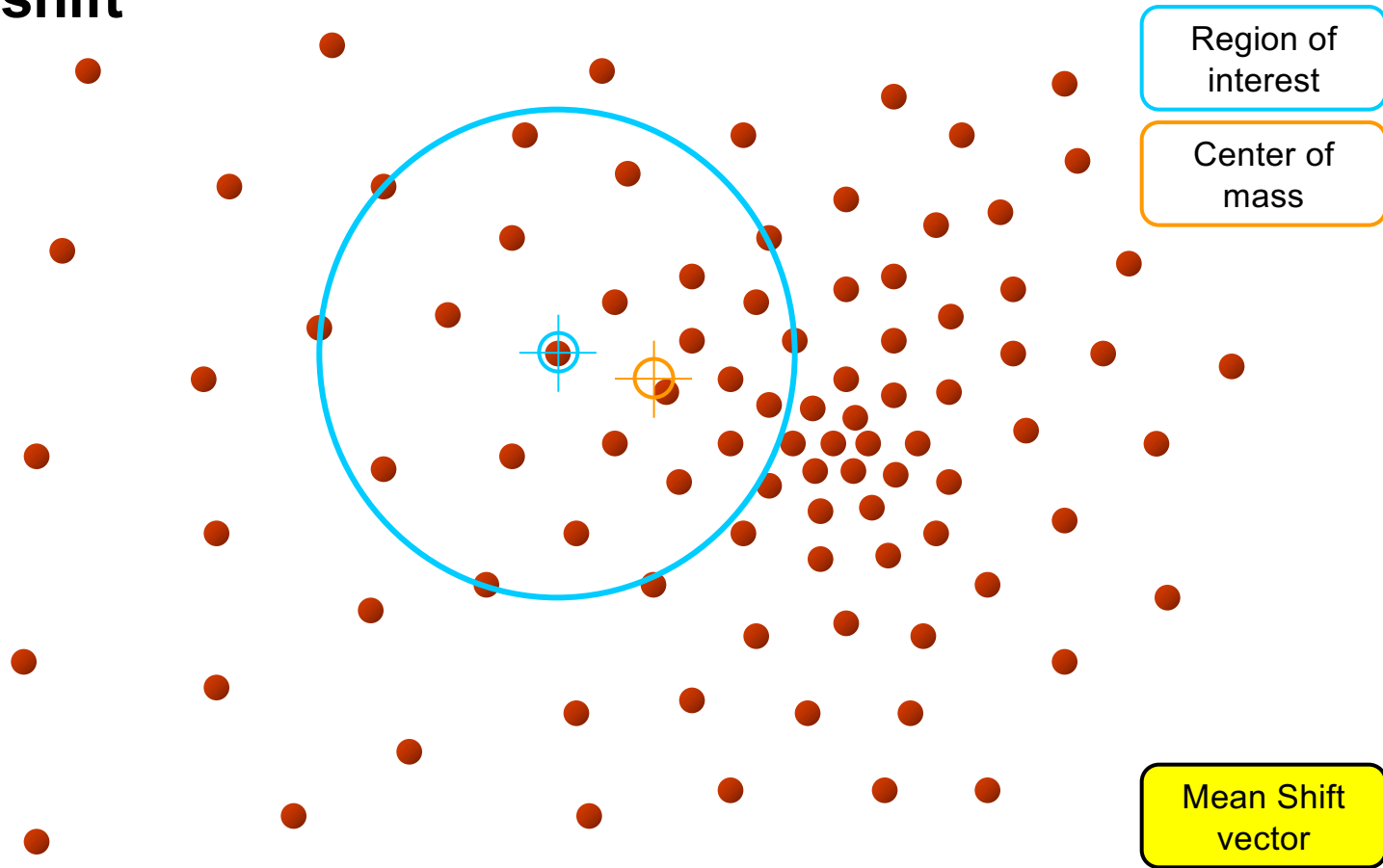
$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right)$$

Gaussian kernel (typically used)

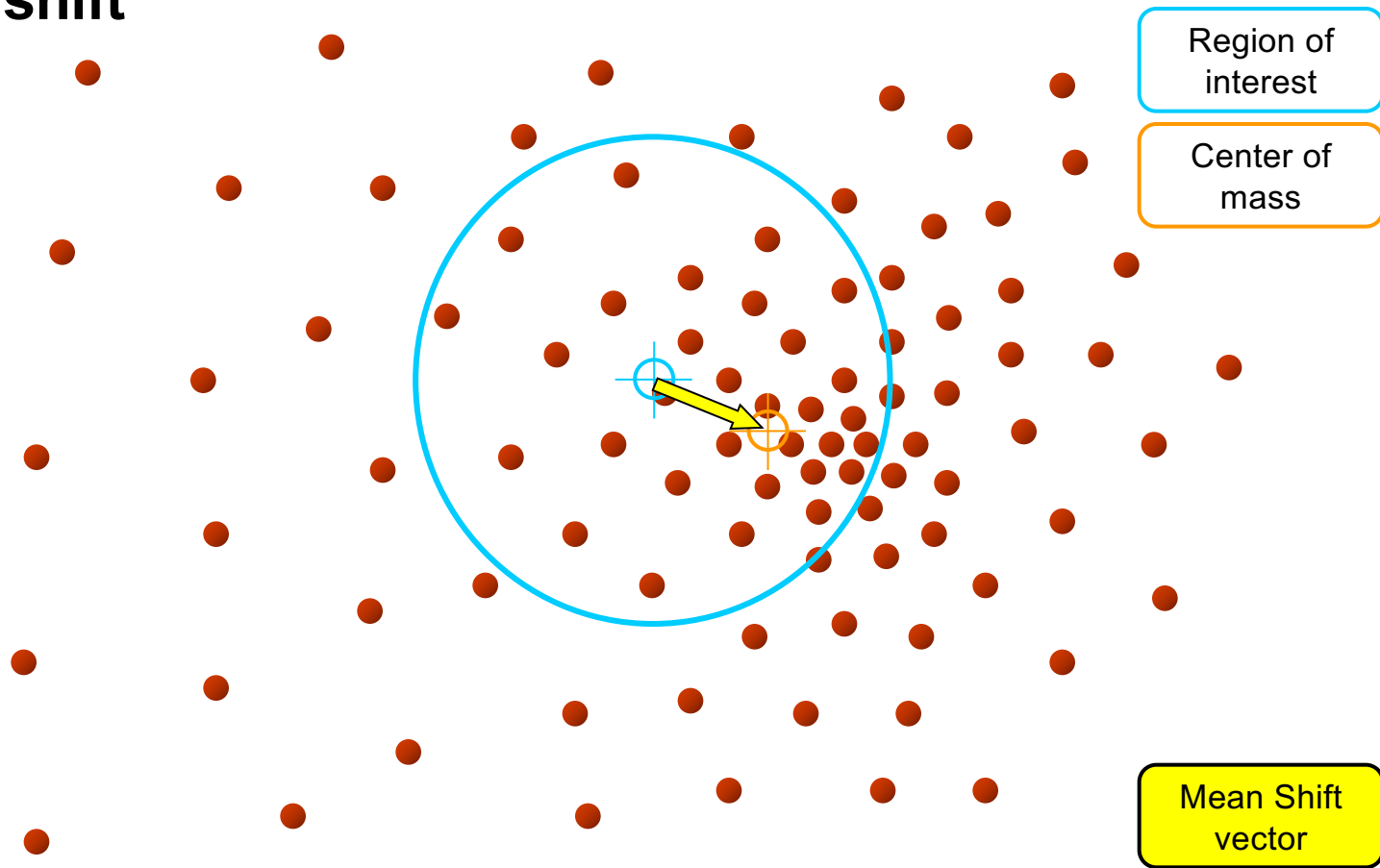
$$K\left(\frac{x - x_i}{h}\right) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x - x_i)^2}{2h^2}}.$$

Mean shift

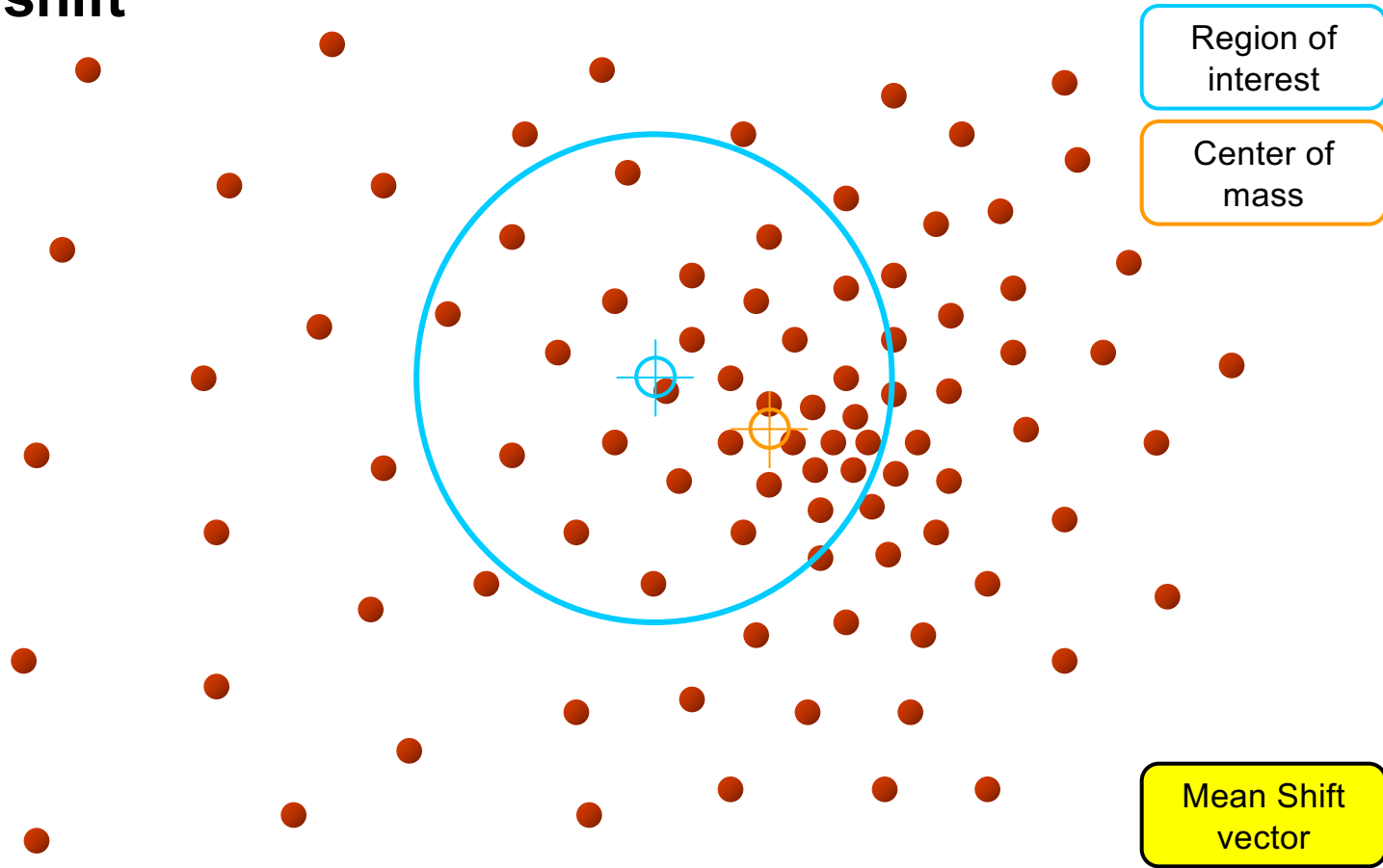


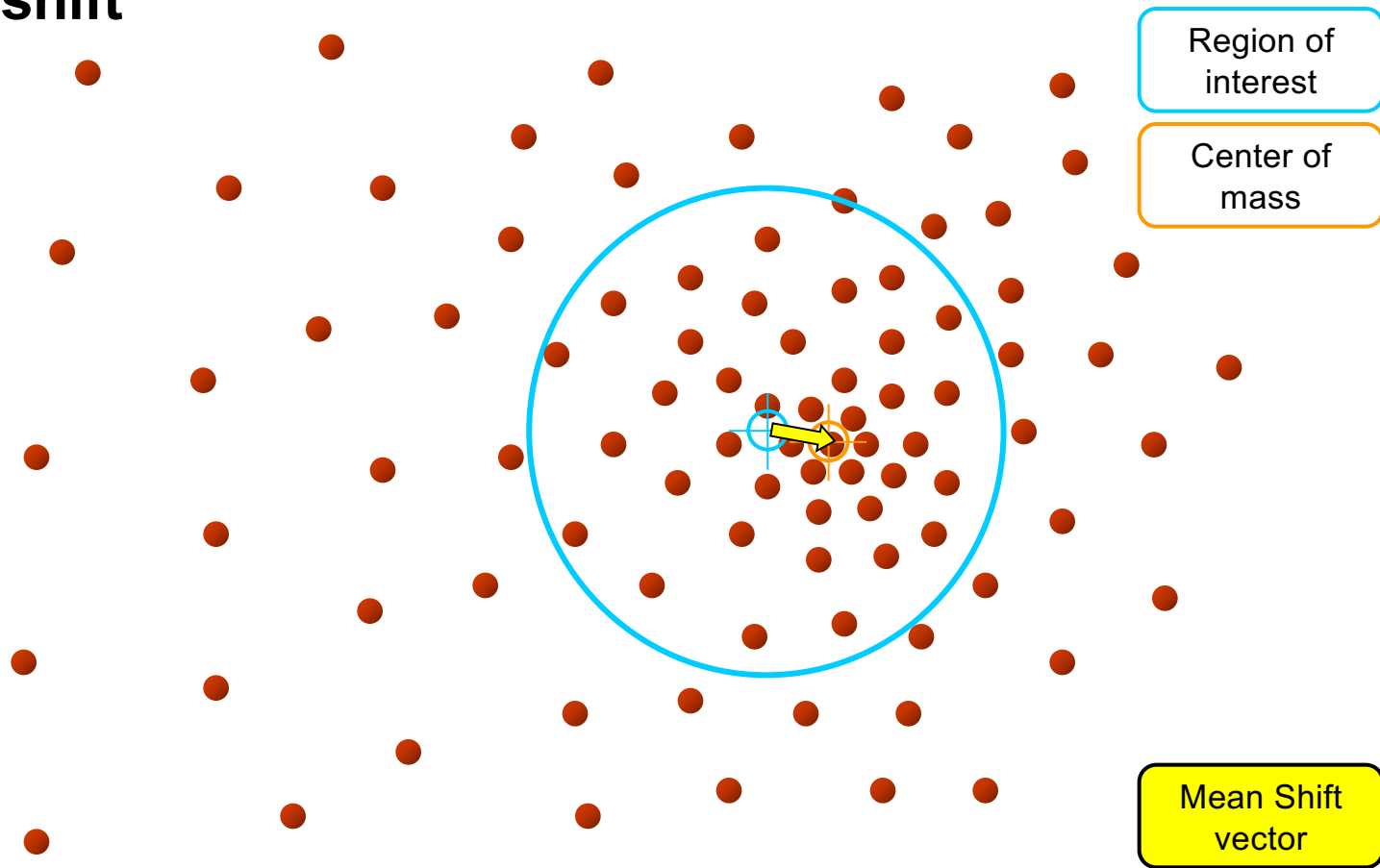


Mean shift

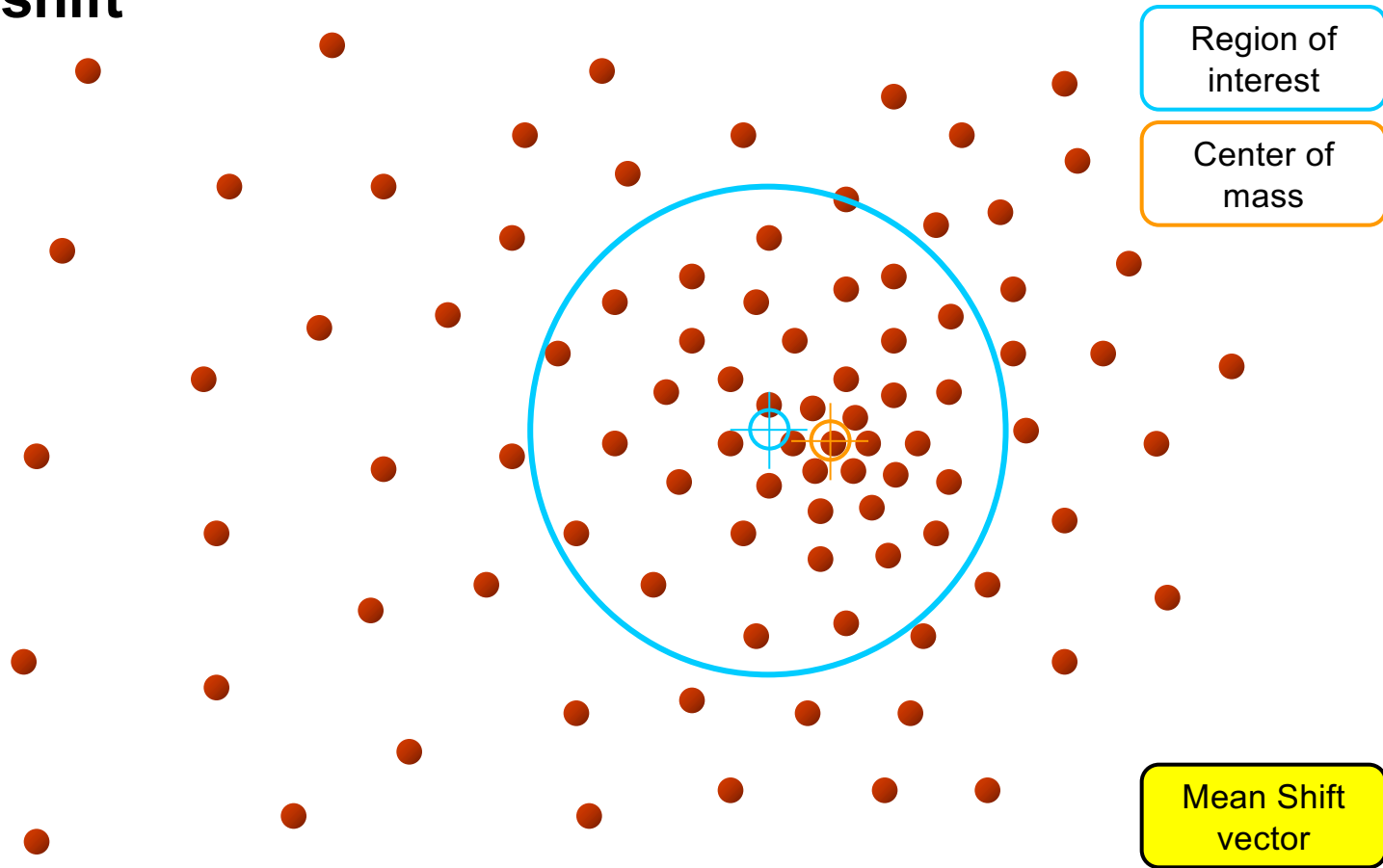


Mean shift

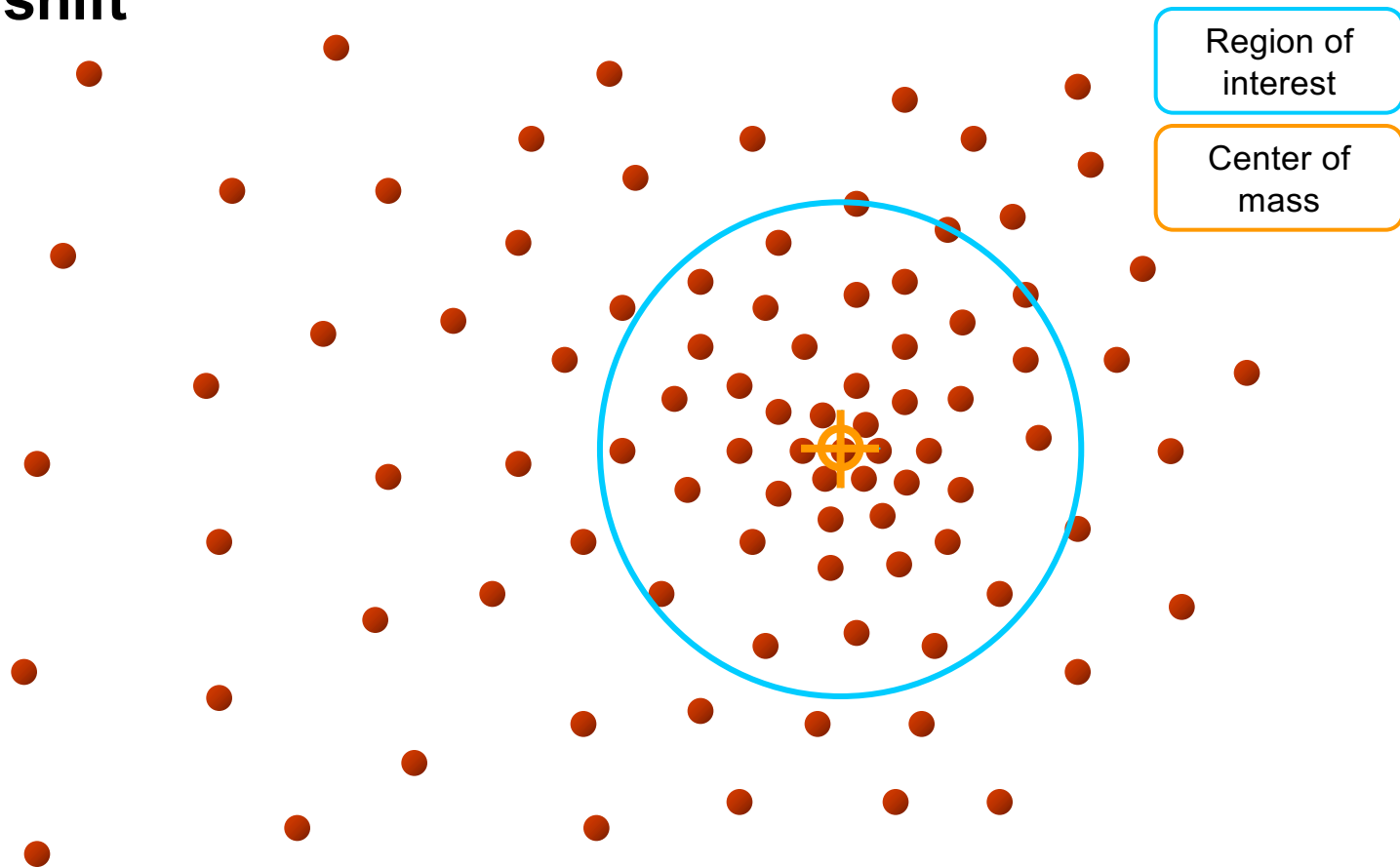




Mean shift

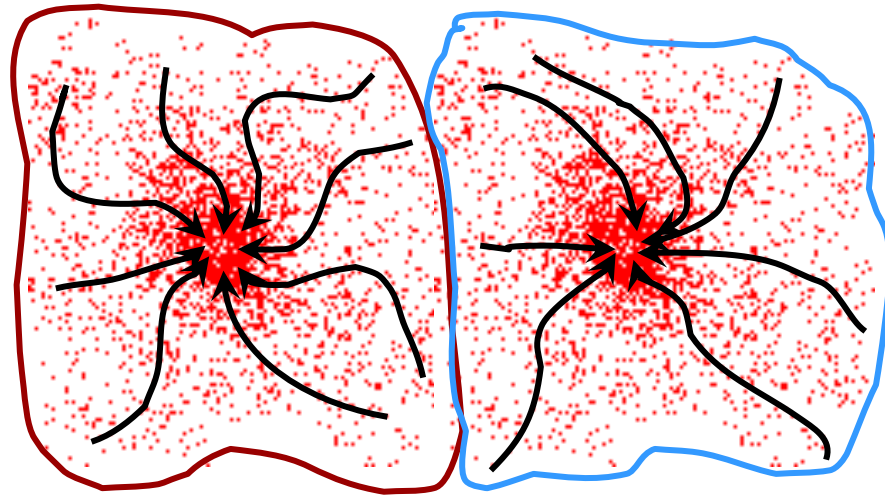


Mean shift

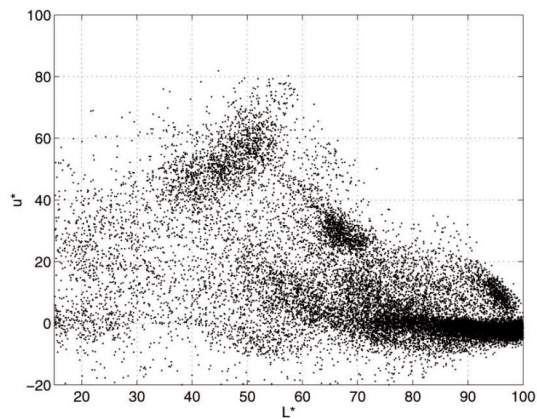


Attraction basin

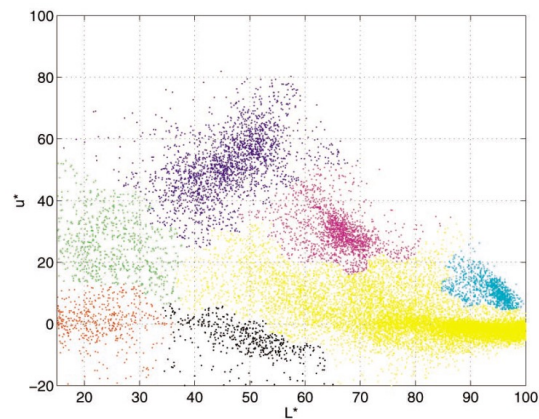
- **Attraction basin:** the region for which all trajectories lead to the same mode
- **Cluster:** all data points in the attraction basin of a mode



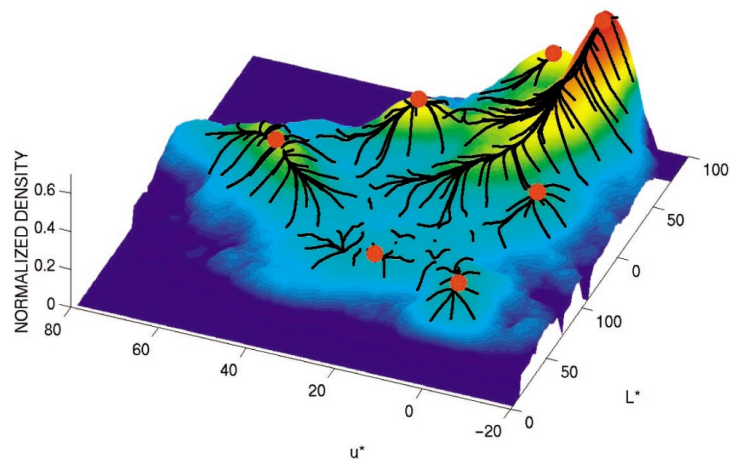
Attraction basin



(a)



(b)



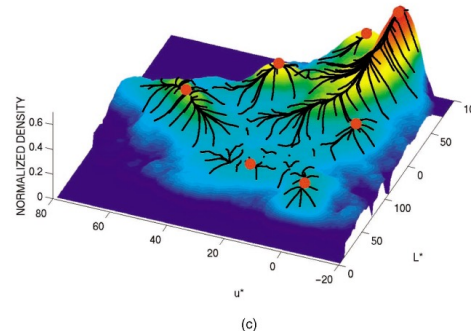
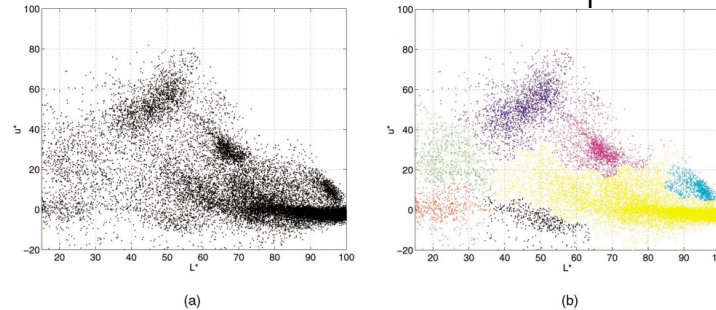
Mean shift clustering

The mean shift algorithm seeks *modes* of the given set of points

1. Choose kernel and bandwidth
2. For each point:
 - a) Center a window on that point
 - b) Compute the mean of the data in the search window
 - c) Center the search window at the new mean location
 - d) Repeat (b,c) until convergence
3. Assign points that lead to nearby modes to the same cluster

2D Image Segmentation by Mean Shift

- Compute features for each pixel (color, gradients, texture, etc)
- Set kernel size for features K_f and position K_s
- Initialize windows at individual pixel locations
- Perform mean shift for each window until convergence
- Merge windows that are within width of K_f and K_s



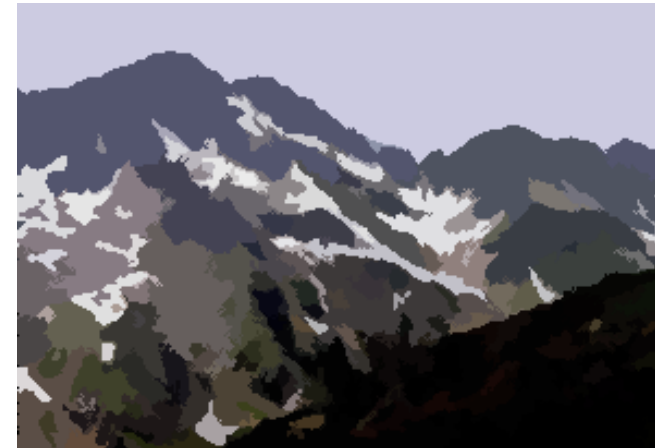
3D Point Cloud Segmentation by Mean Shift

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What should be the dimensionality?

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- dimensionality = 6 ? $\rightarrow$ X, Y, Z, L, U, V
- dimensionality = 6 ? $\rightarrow$ X, Y, Z, H, S, V
- dimensionality > 6 ? $\rightarrow$ X, Y, Z, H, S, V, gradients, texture, ... ??

Mean shift segmentation results



Clustering - DBSCAN

- DBSCAN is a density-based clustering algorithm
- In density-based clustering:
 - we partition points into dense regions separated by not-so-dense regions.
 - a cluster is defined as a maximal set of density-connected points
 - we can discover clusters of arbitrary shape
- Density definition:
 - Density at point p is defined as the number of points within a circle/sphere of radius ϵ
 - A region is dense if the circle/sphere of radius ϵ contains at least *MinPts* points

M. Ester, H.-P. Kriegel, J. Sander, and X. Xu, "A density-based algorithm for discovering clusters in large spatial databases with noise," in International Conference on Knowledge Discovery and Data Mining, 1996.

Clustering - DBSCAN

Types of points:

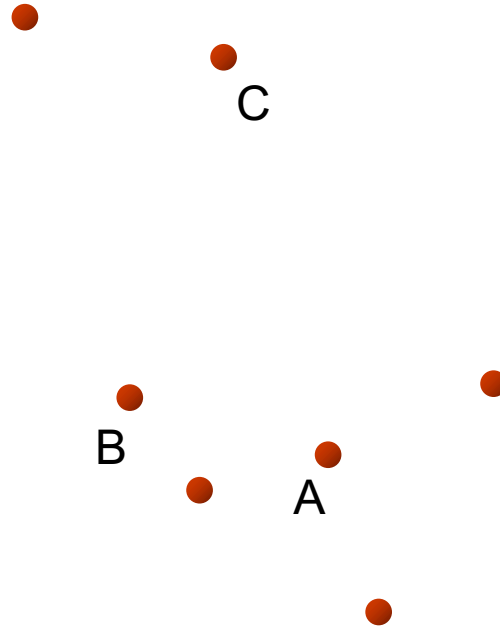
- A **core point** has more than a specified number of points (*MinPts*) within ϵ
- A **border point** has fewer than *MinPts* within ϵ , but is in the neighborhood of a core point.
- A **noise point** is any point that is not a core point or a border point.

Clustering - DBSCAN

Assume:

$$\varepsilon=1$$

$$\text{MinPts}=4$$



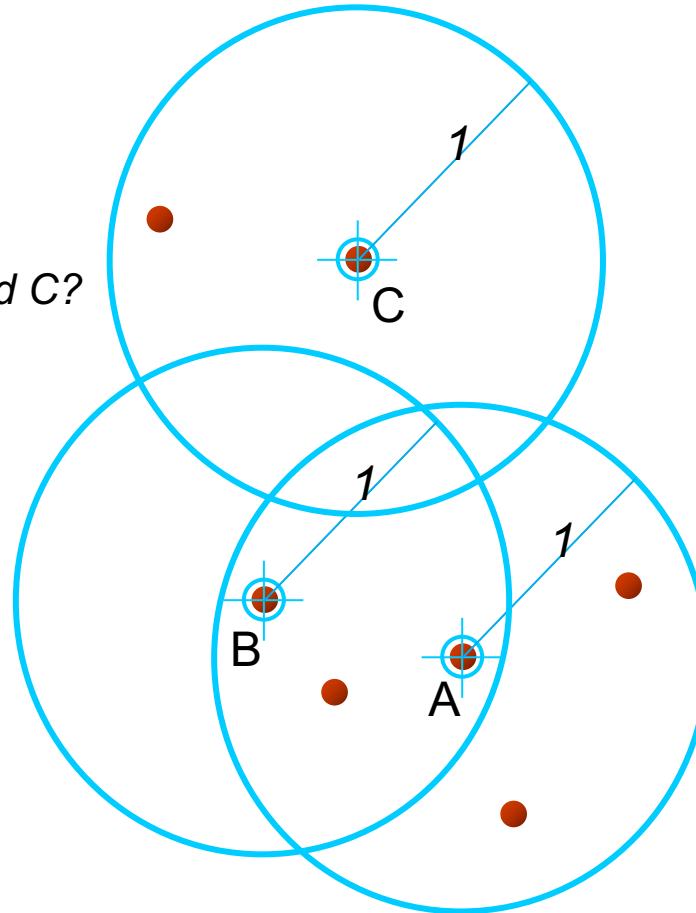
Clustering - DBSCAN

Assume:

$$\epsilon=1$$

$MinPts=4$

What is the type of points A, B and C?

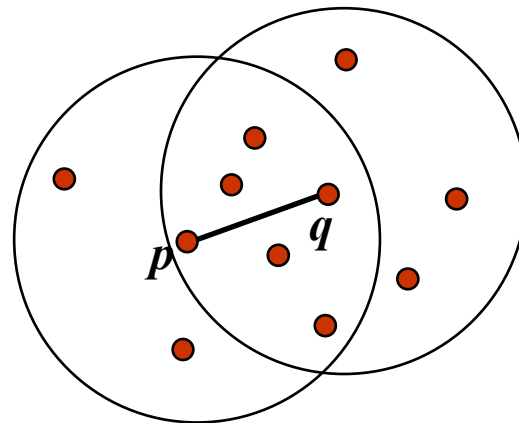


Clustering - DBSCAN

Some more definitions:

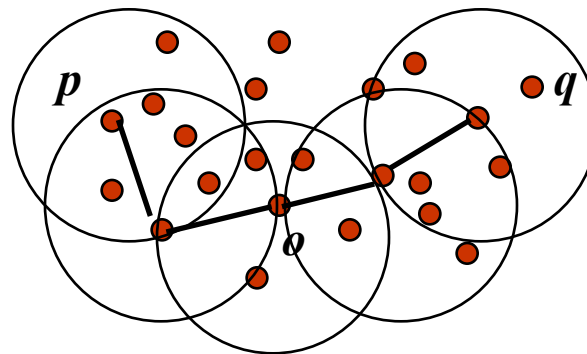
- **Density edge**

- We place an edge between two core points q and p if they are within distance ε



- **Density-connected**

- A point p is density-connected to a point q if there is a path of edges from p to q



Clustering - DBSCAN

DBSCAN algorithm

1. Label points as core, border and noise
2. Eliminate noise points
3. For every core point p that has not been assigned to a cluster
 - Create a new cluster with the point p and all the points that are density-connected to p .
4. Assign border points to the cluster of the closest core point.

A cluster contains:

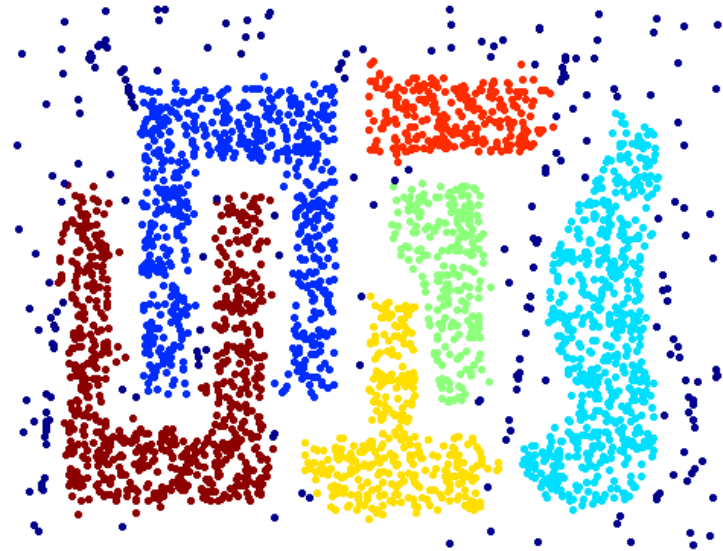
- all core points that can be reached by following a sequence of density-connected core points
- border points that are closest to one of the above-clustered core points

Clustering - DBSCAN

Original Points



Clusters



Clustering - DBSCAN

Pros

- A cluster can have non-convex shape
- No need to define the number of clusters
- Resistant to noise

Cons

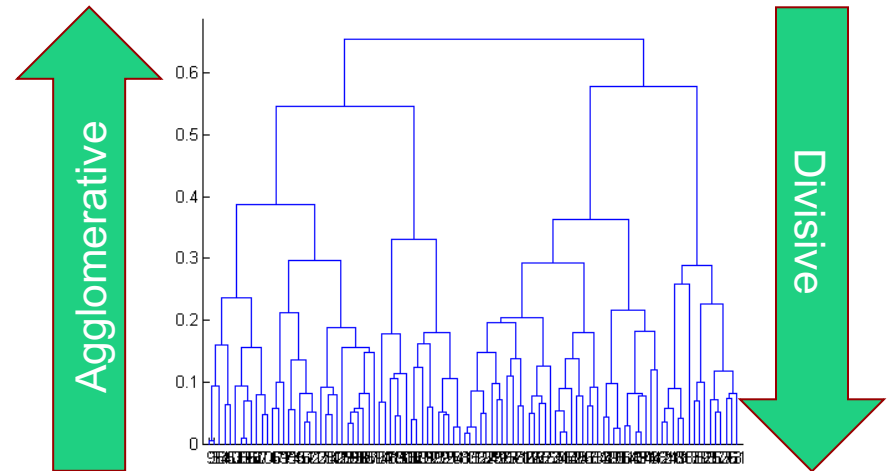
- Some points (noise points) are not assigned to any cluster (is this necessarily bad?)
- Need to define ε and *MinPts*
- Problems with point clouds that contain regions of varying densities

Clustering – Hierarchical Clustering

Builds a **hierarchy of clusters**, *i.e.*, clusters that consist of other smaller clusters.

The 2 types of hierarchical clustering:

- **Agglomerative**: This is a "**bottom-up**" approach. *Each point starts in its own cluster, and pairs of clusters are merged as one moves up the hierarchy.*
- **Divisive**: This is a "**top-down**" approach. *All points start in one cluster, and splits are performed recursively as one moves down the hierarchy.*



Agglomerative / Bottom-up hierarchical clustering

is based only on distance (similarity) between the points.

Algorithm steps:

1. Start by assigning each point to its own cluster, obtaining as many clusters as points.
2. Find the closest (most similar) pair of clusters and merge them into a single cluster.
3. Compute distances (similarities) between the new cluster and each of the old clusters.
4. Repeat steps 2 and 3 until all items are clustered into a single cluster.

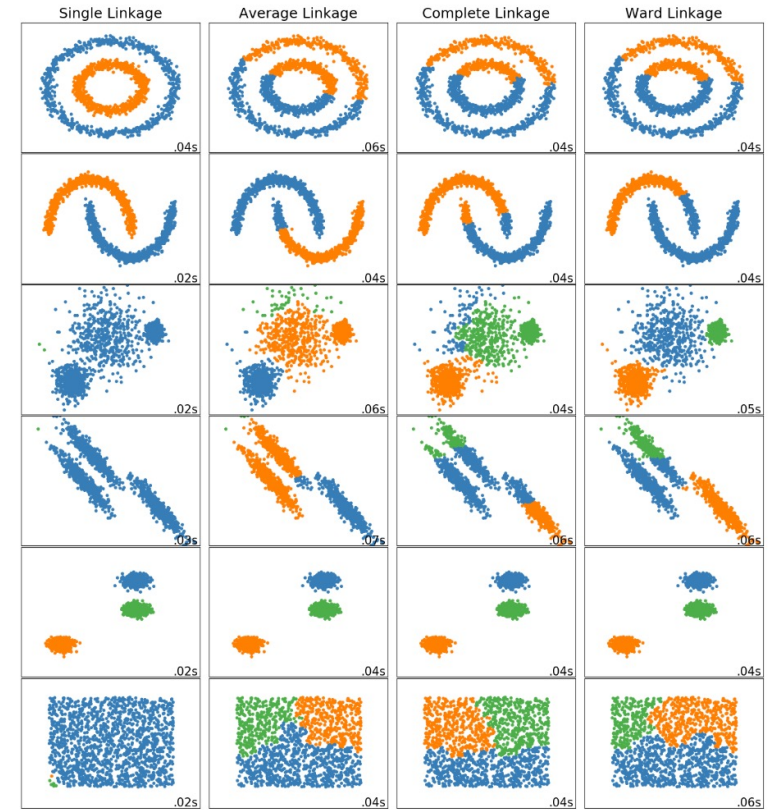
Agglomerative clustering is much more popular than Divisive hierarchical clustering.

DTU Clustering – Hierarchical Clustering

How is distance (similarity) between clusters assessed?

Linkage methods:

- **Single-link:** the distance between two clusters is equal to the minimum distance from any member of one cluster to any member of the other cluster.
- **Complete-link:** the distance between two clusters is equal to the maximum distance from any member of one cluster to any member of the other cluster.
- **Average link:** the distance between two clusters is equal to the average distance from any member of one cluster to any member of the other cluster.
- **Ward-link:** the distance between two clusters is equal to the sum of squared differences within any member of one cluster to any member of the other cluster.





Clustering – Hierarchical Clustering

Hierarchical clustering representation: The output of the hierarchical clustering is a dendrogram.

A dendrogram:

- Consists of many Π -shaped lines that connect data points in a hierarchical tree.
- The height of each Π represents the distance between the two data points being connected.

Characteristics of Hierarchical Clustering:

- Any desired number of clusters can be obtained by ‘cutting’ the dendrogram at the proper level
- They may correspond to meaningful taxonomies



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- Machine Learning techniques can provide useful tools for:
 - finding the underlying structure or
 - fitting modelsto point clouds.
- Regression can fit models of lines, planes, non-linear surfaces
- Clustering is a very powerful tool.
 - No need for labeled data, as it belongs to the unsupervised learning algorithms
 - Many different methods for clustering with different pros and cons.
- 3D point clouds might need to be expanded in higher dimensional spaces to consider additional information (e.g. color, gradients, texture, ...)
 - Machine Learning techniques still applicable in these higher dimensional spaces!

Lazaros Nalpantidis

3D Point Cloud Processing - Clustering & Regression